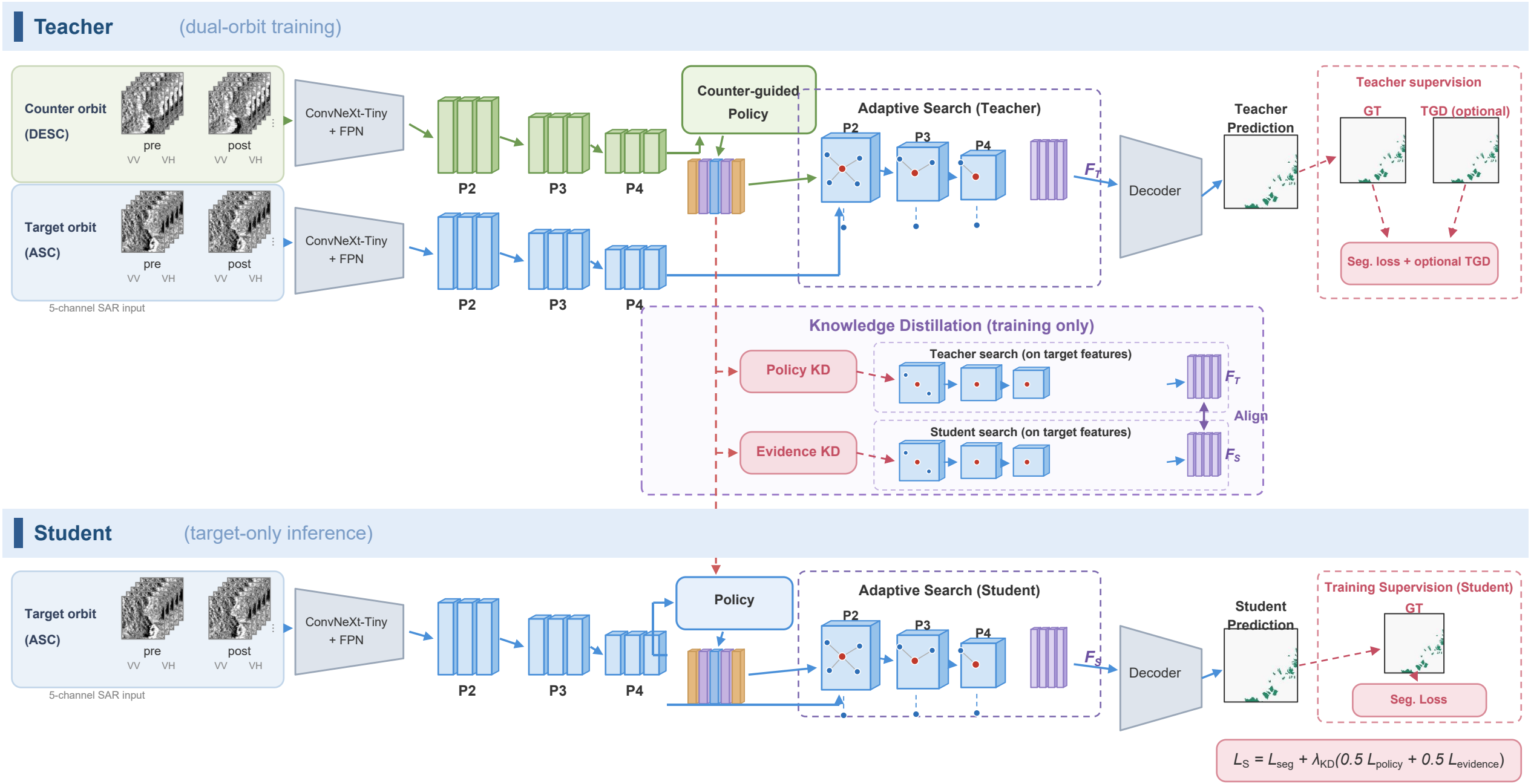

Teacher
(dual-orbit training)
Counter orbit
(DESC)
pre
post
VV
VH
Target orbit
(ASC)
5-channel SAR input
ConvNeXt-Tiny
+ FPN
P2
P3
P4
Counter-guided
Policy
Adaptive Search (Teacher)
F_T
Decoder
Teacher
Prediction
Teacher supervision
GT
TGD (optional)
Seg. loss + optional TGD
Knowledge Distillation (training only)
Policy KD
Teacher search (on target features)
Evidence KD
Student search (on target features)
Align
F_S
Student
(target-only inference)
Policy
Adaptive Search (Student)
Student
Prediction
Training Supervision (Student)
Seg. Loss
$L_S = L_{seg} + \lambda_{KD}(0.5\,L_{policy} + 0.5\,L_{evidence})$